

IoT-Enabled Smart Agriculture: Architecture, Applications, and Challenges

This paper reviews the complete architecture, applications, and challenges of IoT-enabled smart agriculture. The main objective of the paper is to show how IoT can improve agricultural productivity, reduce manual labor, support data-based decisions, and improve farming efficiency. The paper explains that smart agriculture uses sensors, communication technologies, cloud computing, big data, and user applications to monitor and control farming activities.

The paper discusses three main parts of an IoT smart agriculture system: IoT devices, communication technologies, and data processing/storage. IoT devices include sensors, actuators, microcontrollers, communication modules, memory, input-output interfaces, and battery or energy modules. Sensors collect data such as soil moisture, temperature, humidity, pH, water quality, light, and crop condition. Actuators perform actions such as irrigation, spraying, or environmental control. Communication technologies include Wi-Fi, Bluetooth, ZigBee, LoRa, Sigfox, NB-IoT, and other wireless systems. Cloud storage and big data analytics are used to store and process farm data.

The paper includes several architecture diagrams. One diagram shows an IoT ecosystem for smart agriculture, where sensors collect farm data and send it through communication technology to cloud storage and data processing. Another diagram shows the common architecture of an IoT device, including processor, communication module, memory, energy module, sensors, and actuators. The paper also shows a smart agriculture topology where sensor nodes send data to backhaul nodes, then to a controller, data center, or farmer application.

The system works by collecting data from the farm environment. Sensors placed in the field measure soil moisture, air temperature, humidity, water quality, light, and other parameters. These sensors send the collected data to a gateway or backhaul node. From there, the data goes to a cloud server or database. The cloud system stores and analyzes the data using big data or AI-based methods. After analysis, the system can give recommendations to the farmer or control actuators automatically. The farmer can monitor the system using a mobile or web application.

This paper is strongly aligned with our AI-IoT smart farming rover because it provides the complete architecture for our project. Our rover will work like a mobile IoT device. It will carry sensors, collect data from the field, send data through Wi-Fi or another communication module, and display the result on a dashboard. The paper also supports the use of AI and data analytics, which can help our rover detect crop problems and suggest actions. The precision farming concept from this paper is also related to our rover because our robot will move in the field and collect location-based farming data.

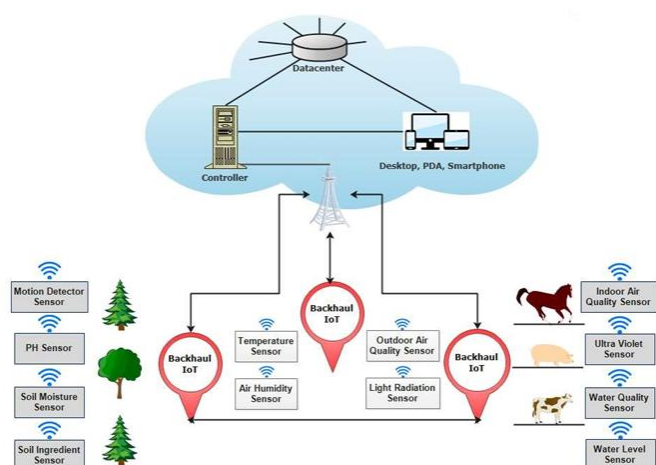


Figure 5. An illustration of the common IoT-based smart agriculture topology.

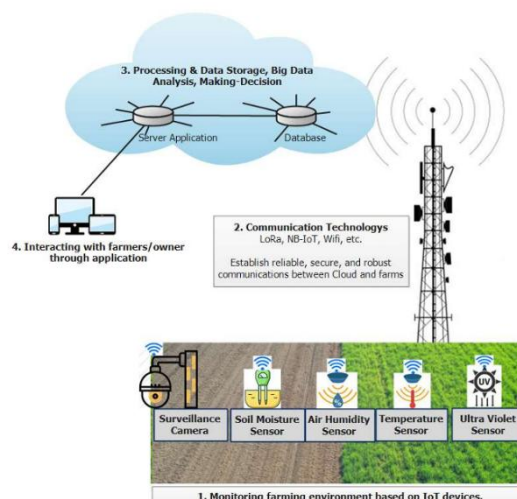


Figure 2. An illustration of IoT ecosystems' architecture for smart agriculture.